

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12407031
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Lee; Jung-Hoon et al.

Sub-pack including multiple unit modules and BMS assembly, and battery pack comprising same

Abstract

Disclosed herein is a sub pack. The sub pack may include a battery module assembly having a plurality of unit modules and a BMS assembly. The battery module assembly may include a side beam. The BMS assembly may be electrically connected to the battery module assembly by wire bonding and include a plurality of sensing terminals at an upper portion thereof. The plurality of sensing terminals may be wire-bonded to the battery module assembly through a connecting wire.

Inventors:	Lee; Jung-Hoon (Daejeon, KR), Yoon; Doo-Han (Daejeon, KR), Kim; Young-Hwan (Daejeon, KR), Yang; Jae-Hun (Daejeon, KR)
Applicant:	LG Energy Solution, Ltd. (Seoul, KR)
Family ID:	76476793
Assignee:	LG Energy Solution, Ltd. (Seoul, KR)
Appl. No.:	17/629187
Filed (or PCT Filed):	December 18, 2020
PCT No.:	PCT/KR2020/018715
PCT Pub. No.:	WO2021/125897
PCT Pub. Date:	June 24, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220255145 A1	Aug. 11, 2022

Foreign Application Priority Data

Publication Classification

Int. Cl.: H01M50/569 (20210101); H01M10/42 (20060101); H01M10/48 (20060101)

U.S. Cl.:

CPC H01M10/425 (20130101); H01M10/48 (20130101); H01M50/569 (20210101); H01M2010/4271 (20130101); H01M2220/20 (20130101)

Field of Classification Search

CPC: H01M (10/425); H01M (10/48); H01M (2010/4271); H01M (2220/20); H01M (50/569)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10243183	12/2018	Harris, III	N/A	N/A
10243184	12/2018	Harris, III	N/A	N/A
10305078	12/2018	Harris, III	N/A	N/A
2005/0260488	12/2004	Zhou	180/68.5	H01M 50/209
2012/0268069	12/2011	Park	320/116	H01M 10/425
2013/0078487	12/2012	Shin et al.	N/A	N/A
2017/0077558	12/2016	Nystrom et al.	N/A	N/A
2017/0179542	12/2016	Cho	N/A	H01M 10/482
2018/0026243	12/2017	Stojanovic et al.	N/A	N/A
2018/0212215	12/2017	Park et al.	N/A	N/A
2019/0081370	12/2018	Capati et al.	N/A	N/A
2019/0202310	12/2018	Gebhart	N/A	N/A
2019/0334216	12/2018	Kim et al.	N/A	N/A
2019/0372069	12/2018	Lee	N/A	H01M 50/262
2020/0006815	12/2019	Hwang et al.	N/A	N/A
2020/0036194	12/2019	Park et al.	N/A	N/A
2020/0058918	12/2019	Shin et al.	N/A	N/A
2023/0010305	12/2022	Lee	N/A	H01M 50/298
2023/0091154	12/2022	Gao	180/65.1	H01M 10/0525

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201508859	12/2009	CN	N/A
102804447	12/2011	CN	N/A
205752417	12/2015	CN	N/A
107925030	12/2017	CN	N/A
110114932	12/2018	CN	N/A
2722909	12/2013	EP	N/A
2018116775	12/2017	JP	N/A
20140008123	12/2013	KR	H01M 50/213
20180091325	12/2017	KR	N/A
20190034868	12/2018	KR	N/A
20190040414	12/2018	KR	N/A
20190052156	12/2018	KR	N/A
20190053125	12/2018	KR	N/A
102029407	12/2018	KR	N/A
20190124446	12/2018	KR	N/A
2012-173233	12/2011	WO	N/A
WO-2015105335	12/2014	WO	H01M 2/105

OTHER PUBLICATIONS

International Search Report for PCT/KR2020/018715 dated Mar. 26, 2021. 3 pgs. cited by applicant

Extended European Search Report including Written Opinion for Application No. 20903278.8 dated Oct. 21, 2022, pp. 1-7. cited by applicant

Search Report dated Aug. 4, 2023 from the Office Action for Chinese Application No. 202080064165.9 issued Aug. 9, 2023, pp. 1-3. [See pp. 1-2, categorizing the cited references]. cited by applicant

Primary Examiner: Laios; Maria

Assistant Examiner: Lu; Ziheng

Attorney, Agent or Firm: Lerner David LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/KR2020/018715 filed on Dec. 18, 2020, which claims priority to Korean Patent Application No. 10-2019-0172440 filed on Dec. 20, 2019, in the Republic of Korea, the disclosures of all of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a sub pack including a plurality of unit modules and a battery management system (“BMS”) assembly, and a battery pack including the same, and more specifically, to a sub pack in which a plurality of unit modules are electrically connected through connecting wires in series and parallel, and a battery pack in which a pair of sub packs are

connected in series through a serial bus bar.

BACKGROUND ART

(3) A secondary battery easily applicable to various product groups and having electrical characteristics such as high energy density is universally applied to not only portable devices but also electric vehicles (EV) or hybrid vehicles (HEV) driven by an electric drive source. The secondary battery is attracting attention as a new energy source for eco-friendliness and energy efficiency because of a primary advantage that it may dramatically reduce the use of fossil fuels and a secondary advantage that it does not generate by-products from the use of energy.

(4) Types of secondary batteries currently widely used in the art include lithium ion batteries, lithium polymer batteries, nickel cadmium batteries, nickel hydride batteries, nickel zinc batteries, and the like. A unit secondary battery cell, namely a unit battery cell, has an operating voltage of about 2.5V to 4.5V. Therefore, if a higher output voltage is demanded, a battery pack is formed by connecting a plurality of battery cells in series. In addition, the battery pack may be formed by connecting a plurality of battery cells in parallel according to the charge/discharge capacity required for the battery pack. Accordingly, the number of battery cells included in the battery pack may be variously set according to the required output voltage or charge/discharge capacity.

(5) Meanwhile, when a battery pack is configured by connecting a plurality of battery cells in series/parallel, generally, a battery module including at least one battery cell is configured first, and other components are added to the at least one battery module to configure a battery pack.

(6) Therefore, it is necessary to minimize a loss in terms of component cost reduction and management by developing a battery pack having a structure in which a plurality of unit modules may be electrically/mechanically coupled conveniently with each other and the unit modules and a battery management system (BMS) may be electrically/mechanically coupled conveniently with each other.

DISCLOSURE

Technical Problem

(7) The present disclosure is designed to solve the problems of the related art, and therefore the present disclosure is directed to minimizing a loss in terms of component cost reduction and management by providing a sub pack having a structure in which a plurality of unit modules may be electrically/mechanically coupled conveniently with each other and the unit modules and a BMS may be electrically/mechanically coupled conveniently with each other, and a battery pack including such sub packs.

(8) However, the technical problem to be solved by the present disclosure is not limited to the above, and other problems not mentioned above may be clearly understood by those skilled in the art from the following description.

Technical Solution

(9) In one aspect of the present disclosure, there is provided a sub pack, comprising: a battery module assembly having a plurality of unit modules; and a BMS assembly coupled to one longitudinal side of the battery module assembly, wherein the battery module assembly includes a side beam provided to at least one side surface of the plurality of unit modules, the BMS assembly is electrically connected to the battery module assembly by wire bonding and has a plurality of sensing terminals provided to an upper portion thereof, and the plurality of sensing terminals are wire-bonded to the battery module assembly through a connecting wire.

(10) The battery module assembly may include a first sub module having a plurality of unit modules connected along a longitudinal direction thereof; and a second sub module having a plurality of unit modules and connected to the first sub module along a width direction of the unit module.

(11) The BMS assembly may include a first BMS configured to control charging and discharging of the first sub module; a second BMS configured to control charging and discharging of the second sub module; and a BMS frame fastened to one longitudinal side of the battery module assembly

while surrounding peripheries of the first BMS and the second BMS.

(12) The unit module may include a plurality of battery cells configured to form a plurality of cell groups; a lower housing configured to support the plurality of battery cells; an upper housing coupled to an upper portion of the lower housing to provide an accommodation space for the plurality of battery cells; and a plurality of parallel bus bars placed on an upper surface of the upper housing and arranged side by side to be spaced apart from each other.

(13) The BMS assembly may include a plurality of BMSs; and a BMS frame configured to surround and fix the plurality of BMSs at once.

(14) The first sub module and the second sub module may be connected to each other in series by an electric connection between parallel bus bars adjacent to each other.

(15) The parallel connection between the plurality of battery cells included in the same cell group, the parallel connection between the plurality of unit modules included in the same sub module and the serial connection between the plurality of sub modules adjacent to each other may be made by wire bonding.

(16) Each of the first BMS and the second BMS may include a plurality of sensing terminals, the sensing terminals of the first BMS may be electrically connected to the parallel bus bar of the first sub module by wire bonding, and the sensing terminals of the second BMS may be electrically connected to the parallel bus bar of the second sub module by wire bonding.

(17) In addition, the present disclosure provides a battery pack in which a pair of sub packs according to the above embodiment are coupled to each other in a mirror-symmetric form based on a central axis parallel to a longitudinal direction of the sub pack.

(18) Moreover, present disclosure provides a vehicle, comprising the battery pack according to the above embodiment.

Advantageous Effects

(19) According to an embodiment of the present disclosure, a plurality of unit modules of a battery pack may be electrically/mechanically coupled conveniently with each other and the unit modules and a BMS may be electrically/mechanically coupled conveniently with each other, and a battery pack including such sub packs, thereby minimizing a loss in terms of component cost reduction and management.

Description

DESCRIPTION OF DRAWINGS

(1) The accompanying drawings illustrate a preferred embodiment of the present disclosure and together with the foregoing disclosure, serve to provide further understanding of the technical features of the present disclosure, and thus, the present disclosure is not construed as being limited to the drawing.

(2) FIG. 1 is a perspective view showing a battery pack according to an embodiment of the present disclosure.

(3) FIG. 2 is a front perspective view of the battery pack depicted in FIG. 1.

(4) FIG. 3 is a perspective view showing a sub pack according to an embodiment of the present disclosure.

(5) FIG. 4 is a perspective view showing a lower surface of the sub pack depicted in FIG. 3.

(6) FIG. 5 is a perspective view showing a battery module assembly of the sub pack of FIG. 3 according to an embodiment of the present disclosure.

(7) FIG. 6 is a perspective view showing a unit module of the battery module assembly depicted in FIG. 5.

(8) FIG. 7 is a diagram showing a lower surface of the unit module depicted in FIG. 6.

(9) FIG. 8 is a partially enlarged view of the battery module assembly depicted in FIG. 5.

(10) FIG. 9 is a partial perspective view showing an upper surface of the unit module depicted in FIG. 6.

(11) FIG. 10 is a diagram showing an electric connection between the unit modules of the battery module assembly depicted in FIG. 5.

(12) FIG. 11 is a diagram showing a rear block of the battery module assembly depicted in FIG. 5.

(13) FIG. 12 is a first perspective view showing a BMS assembly of the sub pack depicted in FIG. 3.

(14) FIG. 13 is a second perspective view of the BMS assembly of FIG. 12.

(15) FIG. 14 is a diagram showing an electric coupling between the BMS assembly of FIG. 12 and the battery module assembly depicted in FIG. 5.

(16) FIG. 15 is a diagram showing an electric connection between a pair of sub packs according to an embodiment of the present disclosure.

(17) FIG. 16 is a diagram showing a coupling structure between a BMS assembly and a battery module assembly the sub pack of FIG. 15 according to an embodiment of the present disclosure.

BEST MODE

(18) Hereinafter, preferred embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. Prior to the description, it should be understood that the terms used in the specification and the appended claims should not be construed as limited to general and dictionary meanings, but interpreted based on the meanings and concepts corresponding to technical aspects of the present disclosure on the basis of the principle that the inventor is allowed to define terms appropriately for the best explanation. Therefore, the description proposed herein is just a preferable example for the purpose of illustrations only, not intended to limit the scope of the disclosure, so it should be understood that other equivalents and modifications could be made thereto without departing from the scope of the disclosure.

(19) Referring to FIGS. 1 to 5, the battery pack according to an embodiment of the present disclosure includes a pair of sub packs 1, a heat sink 2, a serial bus bar 3, and a plurality of fastening bolts 4.

(20) The pair of sub packs 1 are coupled to each other in a mirror-symmetric form based on a central axis parallel to a width direction (Y-axis direction) of the battery pack with the heat sink 2 being interposed therebetween. In other words, the pair of sub packs 1 are coupled in a state where bottom surfaces thereof face each other. Hereinafter, when necessary for convenience of description, the sub pack 1 positioned at an upper portion will be referred to as a first sub pack, and the sub pack 1 positioned at a lower portion will be referred to as a second sub pack. The detailed structure of the sub pack 1 will be described later in detail with reference to FIG. 3.

(21) The heat sink 2 is a component applied for cooling the battery pack, and is interposed between the pair of sub packs 1 so that both sides thereof contact the pair of sub packs 1, respectively.

(22) Meanwhile, the heat sink 2 may be disposed to be in indirect contact with the pair of sub packs 1 through a heat transfer material, instead of being in direct contact with the pair of sub packs 1. For example, it may be possible that the heat transfer material is disposed between the pair of sub packs 1 and the heat sink 2.

(23) The serial bus bar 3 connects the pair of sub packs 1, which are coupled to each other in a mirror-symmetric form, in series. That is, the parallel bus bar 114 located at an outermost part of one side of the first sub pack 1 in a width direction (a direction parallel to the Y-axis) and the parallel bus bar 114 located at an outermost part of one side of the second sub pack 1 in a width direction (a direction parallel to the Y-axis) have different polarities, and the serial bus bar 3 connects the pair of parallel bus bars 114 so that the pair of sub packs 1 are connected in series.

(24) The fastening bolt 4 may be used for fastening the pair of sub packs 1 to each other. In addition, unlike this, the fastening bolt 4 may be used for fastening the unit modules 110 to each other and fastening the battery module assembly 10 and the BMS assembly 20 to each other, and thus the pair of sub packs 1 completely fabricated as above may be coupled by an adhesive layer

interposed between the first sub pack **1** and the heat sink **2** and between the second sub pack **1** and the heat sink **2**. In addition, it is also possible to apply both fastening by bolting and coupling by interposing an adhesive layer in order to maximize the coupling force.

(25) Next, a schematic structure of the sub pack **1** according to an embodiment of the present disclosure will be described with reference to FIGS. **3** to **5**.

(26) Referring to FIGS. **3** to **5**, the sub pack **1** according to an embodiment of the present disclosure includes a battery module assembly **10** and a BMS assembly **20**.

(27) The battery module assembly **10** includes a plurality of sub modules **100**, a rear block **200** and a side beam **300**.

(28) The plurality of sub modules **100** are connected to each other in series, and each sub module **100** includes a plurality of unit modules **110** connected to each other in parallel. In the figures, it is illustrated that one sub module **100** includes two unit modules **110** coupled to each other and one battery module assembly **10** includes two sub modules **100** coupled to each other, but the present disclosure is not limited thereto. That is, one sub module **100** may include three or more unit modules **110** coupled to each other, and one battery module assembly **10** may include three or more sub modules **100** coupled to each other. Meanwhile, for convenience of explanation, the pair of sub modules **100** shown in FIG. **5** will be distinguishably referred to as a first sub module **100** and a second sub module **100**.

(29) The rear block **200** may be a component applied to mechanically fasten the plurality of sub modules **100**. Hereinafter, the rear block **200** will be described in more detail in the following related description.

(30) The side beam **300** may be provided to at least one side surface or both side surfaces of the plurality of unit modules **110**. The side beam **300** may support the plurality of unit modules **110** more stably. In addition, the side beam **300** may buffer external shocks transmitted to the plurality of unit modules **110**, thereby minimizing the shocks transmitted to the plurality of unit modules **110**.

(31) Hereinafter, a specific structure of the unit module **110** will be described with reference to FIGS. **6** to **9**.

(32) Referring to FIGS. **6** to **9**, the unit module **110** includes a plurality of battery cells **111**, a lower housing **112**, an upper housing **113**, and a plurality of parallel bus bars **114**.

(33) As the battery cell **111**, a cylindrical battery cell may be applied, for example. The battery cell **111** may be provided in plural along a longitudinal direction of the unit module **110** (a direction parallel to the X-axis in FIG. **6**) and provided in plural along a width direction of the unit module **110** (a direction parallel to the Y-axis in FIG. **6**). The plurality of battery cells **111** are arranged to stand upright on the lower housing **112**. Each of the plurality of battery cells **111** includes a positive electrode terminal **111a** and a negative electrode terminal **111b** exposed to the outside through an upper surface of the upper housing **113**. When the battery cell **111** is a cylindrical battery cell, a can accommodating an electrode assembly may function as the negative electrode terminal **111b**.

(34) The plurality of battery cells **111** disposed along the longitudinal direction of the unit module **110** are connected in parallel to each other to form one cell group C. In addition, a plurality of cell groups C arranged along the width direction of the unit module **110** are connected in series with each other. Hereinafter, if necessary for convenience of description, two cell groups C adjacent to each other among the plurality of cell groups C will be distinguishably referred as a first cell group and a second cell group.

(35) Referring to FIG. **9**, such a parallel/serial connection is implemented using a plurality of parallel bus bars **114** and connecting wires W. The parallel bus bars **114** are disposed at both sides of each cell group C. The positive electrode terminals **111a** of the battery cells **111** forming the first cell group C are connected to the parallel bus bar **114** located at one side of the first cell group C by wire bonding. The negative electrode terminals **111b** of the battery cells **111** forming the first cell group C are connected to the parallel bus bar **114** located at the other side of the first cell group C

by wire bonding. Accordingly, the plurality of battery cells **111** forming the first cell group C are connected in parallel with each other.

(36) In addition, the positive electrode terminals **111a** of the plurality of battery cells **111** forming the second cell group C adjacent to the first cell group C are connected to the parallel bus bar **114** located at one side of the second cell group, namely the parallel bus bar **114** located at the other side of the first cell group C, by wire bonding. The negative electrode terminals **111b** of the plurality of battery cells **111** forming the second cell group C are connected to the parallel bus bar **114** located at the other side of the second cell group C by wire bonding. Accordingly, the plurality of battery cells **111** forming the second cell group C are connected in parallel with each other, and the first cell group C and the second cell group C are connected in series with each other.

(37) Referring to FIGS. **6** to **9**, the lower housing **112** supports the plurality of battery cells **111**, and may include a fastening protruding portion **112a**, a fixing protruding portion **112b**, a first fastening accommodation portion **112c**, and a first fixing accommodation portion **112d**. The fixing protruding portion **112b** and the first fixing accommodation portion **112d** are not essential components necessarily required, but serve as components for convenience in fastening the unit modules **110** to each other.

(38) At least one fastening protruding portion **112a** is provided at one side of the lower housing **112** in a longitudinal direction (parallel to the X-axis in FIG. **6**), and protrudes outward from the lower housing **112**. The fastening protruding portion **112a** has a bolting hole H1 into which a fastening bolt **4** (see FIG. **1**) may be inserted along a height direction of the unit module **110** (a direction parallel to the Z-axis in FIG. **6**). At least one fixing protruding portion **112b** is provided at one longitudinal side of the lower housing **112** and protrudes outward from the lower housing **112**. The fixing protruding portion **112b** has a fixing protrusion f protruding in an upward direction (a direction parallel to the Z-axis in FIG. **6**).

(39) The first fastening accommodation portion **112c** is provided at the other longitudinal side of the lower housing **112**, and has a shape corresponding to the fastening protruding portion **112a** at a location corresponding thereto. Therefore, when the plurality of unit modules **110** are fastened along the longitudinal direction to form the sub module **100**, the fastening protruding portion **112a** of one unit module **110** is inserted into the first fastening accommodation portion **112c** of another adjacent unit module **110**. In addition, the upper housing **113** has a bolting hole H2 formed at a location corresponding to the first fastening accommodation portion **112c**. Accordingly, when the pair of unit modules **110** are coupled, the bolting hole H1 provided to the lower housing **112** and the bolting hole H2 provided to the upper housing **113** are placed at the same location so that one fastening bolt **4** (see FIG. **1**) may pass therethrough.

(40) The first fixing accommodation portion **112d** is provided at the other longitudinal side of the lower housing **112** and has a shape corresponding to the fixing protruding portion **112b** at a location corresponding thereto. Therefore, when the plurality of unit modules **110** are fastened along the longitudinal direction to form the sub module **100**, the fixing protruding portion **112b** of one unit module **110** is inserted into the first fixing accommodation portion **112d** of another adjacent unit module **110**. In addition, the upper housing **113** has a protrusion accommodating groove G formed at a location corresponding to the first fixing accommodation portion **112d**. Accordingly, when the pair of unit modules **110** are coupled, the fixing protrusion f provided to the lower housing **112** is inserted into the protrusion accommodating groove G provided to the upper housing **113**. By doing so, the pair of unit modules **110** may be fixed not to move in the horizontal direction (a direction parallel to the X-Y plane in FIG. **6**).

(41) The upper housing **113** provides a space for accommodating the plurality of battery cells **111** and has at least one bolting hole H2 and a plurality of cell exposing portions **113a** formed at the upper surface thereof.

(42) As mentioned above, the bolting hole H2 is formed at a location corresponding to the first fastening accommodation portion **112c** of the lower housing **112** to provide a space into which the

fastening bolt **4** (see FIG. **1**) may be inserted.

(43) The cell exposing portion **113a** has a shape extending along a longitudinal direction of the upper housing **113** (a direction parallel to the X-axis in FIG. **6**), and is provided at a location corresponding to the cell group C in the same number as the number of cell groups C. The cell exposing portion **113a** includes a plurality of slits S so that the positive electrode terminal **111a** and the negative electrode terminal **111b** of the battery cell **111** may be exposed to the outside through the upper surface of the upper housing **113**.

(44) In the plurality of the battery cells **111** provided in one unit module **110**, the positive electrode terminal **111a** and the negative electrode terminal **111b** exposed through the slits S are connected to the parallel bus bar **114** by wire bonding to make an electric connection where serial connections and parallel connections are mixed.

(45) The parallel bus bar **114** has a shape extending in a direction parallel to the cell exposing portion **113a**, and is placed on the upper surface of the upper housing **113**. The parallel bus bar **114** is disposed at both sides of each of the plurality of cell exposing portions **113a**. Among the plurality of parallel bus bars **114**, the parallel bus bar **114** disposed at one end of the unit module **110** in a width direction (a direction parallel to the Y-axis in FIG. **6**) and the parallel bus bar **114** disposed at the other end thereof have different polarities. This is to allow the plurality of sub modules **100** to be connected to each other in series along the width direction of the unit module **110**. Referring to FIG. **10**, a pair of parallel bus bars **114** provided at the outermost sides of the sub modules **100** adjacent to each other contact each other or are connected by wire bonding, so that the adjacent sub modules **100** may be connected in series.

(46) Referring to FIG. **11**, the rear block **200** includes a plurality of rear fastening portions **210** and a plurality of rear fixing portions **220**.

(47) The rear fastening portion **210** has a shape corresponding to the first fastening accommodation portion **112c** of the lower housing **112** at a location corresponding thereto. The rear fastening portion **210** has a bolting hole H3 formed in a size corresponding to the bolting hole H2 of the upper housing **113** at a location corresponding thereto.

(48) The rear fixing portion **220** has a shape corresponding to the first fixing accommodation portion **112d** of the lower housing **112** at a location corresponding thereto. Although not shown in the figures, the rear fixing portion **220** may have a fixing protrusion formed in a size corresponding to the protrusion accommodating groove G (see FIG. **7**) of the upper housing **113** at a location corresponding thereto.

(49) Next, the BMS assembly **20** will be described in detail with reference to FIGS. **12** to **16**.

(50) Referring to FIGS. **12** to **16**, the BMS assembly **20** includes a plurality of battery management systems (BMSs) **21** and a BMS frame **22** for surrounding and fixing the plurality of BMS **21** at once. The BMS assembly **20** is coupled to one side of the battery module assembly **10** (see FIG. **5**) in a longitudinal direction (a direction parallel to the X-axis in FIG. **5**). That is, the BMS assembly **20** is provided in the same number as the number of sub packs **1**.

(51) The BMS **21** is provided in plural, and the BMSs **21** are connected to the sub modules **100** in one-to-one relationship through a plurality of sensing terminals T provided to an upper portion thereof. That is, the BMSs **21** are provided in the same number as the number of sub modules **100**. As shown in FIG. **14**, the sensing terminals T of the BMS **21** are connected to the parallel bus bars **114** in one-to-one relationship by wire bonding using a connecting wire W. Accordingly, one BMS **21** may sense voltage and/or current of a plurality of cell groups C (refer to FIG. **7**) and accordingly controls charging and discharging of the sub module **100**. In the figures, it is exemplarily illustrated that two BMSs **21** are provided to each BMS assembly **20**. In this case, the BMSs **21** will be referred to as a first BMS **21** and a second BMS **21** for convenience of explanation.

(52) The first BMS **21** controls charging and discharging of the first sub module **100** (see FIG. **5**), and the second BMS **21** controls charging and discharging of the second sub module **100** (see FIG. **5**) adjacent to the first sub module **100**.

(53) The BMS frame **22** surrounds and fixes the plurality of BMSs **21** at once, and is coupled to one longitudinal side of the battery module assembly **10** (see FIG. 5). The BMS frame **22** has a shape extending along the width direction of the battery module assembly **10** (see FIG. 5) (a direction parallel to the Y-axis in FIG. 5), and has a length corresponding to the width of the battery module assembly **10**.

(54) The BMS frame **22** may include a plurality of front fastening portions **22a**, a plurality of second fastening accommodation portions **22b**, a plurality of second fixing accommodation portions **22c**, and a bus bar placing portion **22d**.

(55) The front fastening portion **22a** is provided at a location corresponding to the fastening protruding portion **112a** and has a bolting hole **H4**. The bolting hole **H4** formed at the front fastening portion **22a** has a shape corresponding to the bolting hole **H1** formed in the lower housing **112** at a location corresponding thereto. Accordingly, one fastening bolt **4** (see FIGS. 1 and 2) may simultaneously pass through the bolting hole **H4** formed in the front fastening portion **22a** and the bolting hole **H1** formed in the lower housing **112**, and thus the battery module assembly **10** and the BMS assembly **20** may be fastened.

(56) The second fastening accommodation portion **22b** has a groove shape at locations corresponding to the front fastening portion **22a** and the fastening protruding portion **112a**. The second fastening accommodation portion **22b** has a shape corresponding to the fastening protruding portion **112a**, and accordingly the fastening protruding portion **112a** may be inserted into the second fastening accommodation portion **22b** so that the BMS assembly **20** and the battery module assembly **10** come into close contact with each other.

(57) The second fixing accommodation portion **22c** has a groove shape at a location corresponding to the fixing protruding portion **112b**. The second fixing accommodation portion **22c** has a shape corresponding to the fixing protruding portion **112b**, and accordingly the fixing protruding portion **112b** may be inserted into the second fixing accommodation portion **22c** so that the BMS assembly **20** and the battery module assembly **10** come into close contact with each other.

(58) The bus bar placing portion **22d** has a groove shape at one longitudinal side of the BMS frame **22**. The bus bar placing portion **22d** provides a space in which the pair of sub packs **1** coupled to each other in a mirror-symmetric form and the serial bus bar **3** for connecting the pair of sub packs **1** in series may be placed. As shown in FIG. 15, the pair of BMS assemblies **20** are coupled to each other in a mirror-symmetric form based on a central axis parallel to the width direction of the battery pack (a direction parallel to the Y-axis in FIG. 15). The bus bar placing portions **22d** respectively provided to the pair of BMS assemblies **20** coupled to each other as above are connected to each other to provide a space in which the serial bus bar **3** may be placed.

(59) As described above, the serial bus bar **3** placed on the bus bar placing portion **22d** connects the parallel bus bars **114** respectively provided at the outermost parts of the pair of sub packs **1** to each other, and accordingly the pair of sub packs **1** are connected to each other in series. The serial bus bar **3** and the parallel bus bar **114** are electrically connected by wire bonding using a connecting wire **W**.

(60) As described above, the battery pack according to an embodiment of the present disclosure not only has a structure in which the plurality of unit modules **110** may be simply fastened by bolting, but also has a structure in which the battery module assembly **10** formed by coupling the plurality of unit modules **110** and the BMS assembly **20** may also be easily fastened by bolting.

(61) That is, as shown in FIGS. 1 and 2, the battery pack according to an embodiment of the present disclosure has a structure in which the fastening bolts **4** are inserted into the bolting holes **H1** to **H4** in a state where the pair of sub packs **1** face each other in a mirror-symmetric form, so that the pair of sub packs **1** are fastened with each other. In addition, the battery pack according to an embodiment of the present disclosure has a structure in which the unit modules **110** of each sub pack **1** may be fastened and the battery module assembly **10** and the BMS assembly **20** may also be fastened simultaneously by fastening the pair of sub packs **1**.

(62) In addition, if the battery pack according to an embodiment of the present disclosure is used, by using the parallel bus bar **114** and the serial bus bar **3** and applying wire bonding to electrically connect components, the plurality of unit modules **110** may be simply connected in series and parallel in a mixed form, and accordingly it is very easy to expand the capacity and output voltage as needed.

(63) Meanwhile, a vehicle according to an embodiment of the present disclosure includes the battery pack according to an embodiment of the present disclosure as described above.

(64) The present disclosure has been described in detail. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the disclosure, are given by way of illustration only, since various changes and modifications within the scope of the disclosure will become apparent to those skilled in the art from this detailed description.

Claims

1. A sub pack comprising: a battery module assembly having a plurality of unit modules, a first unit module of the battery module having a first fastening portion protruding in a first direction, a second unit module of the battery module positioned behind the first unit module and having a second fastening portion protruding in a second direction, a fastening bolt of the battery module traversing the first and second fastening portions; and a battery management system (BMS) assembly coupled to one longitudinal side of the battery module assembly, wherein the battery module assembly includes a side beam provided at least one side surface of the plurality of unit modules, the BMS assembly being electrically connected to the battery module assembly by wire bonding, the BMS assembly including a plurality of sensing terminals provided on an upper surface of the BMS assembly, the plurality of sensing terminals arranged in a vertically tiered configuration, each sensing terminal being wire-bonded directly to a corresponding parallel bus bar of the battery module assembly through a connecting wires.
2. The sub pack according to claim 1, wherein the battery module assembly includes a first sub module and a second sub module, the first sub module having a plurality of unit modules connected along a longitudinal direction thereof, the second sub module having a plurality of unit modules connected to the first sub module along a width direction of the unit module.
3. The sub pack according to claim 2, wherein the BMS assembly includes a first BMS, a second BMS, and a BMS frame, the first BMS configured to control charging and discharging of the first sub module, the second BMS configured to control charging and discharging of the second sub module, the BMS frame being connected to one longitudinal side of the battery module assembly adjacent the first BMS and the second BMS.
4. The sub pack according to claim 2, wherein any one of the plurality of unit modules including: a plurality of battery cells forming a plurality of cell groups; a lower housing supporting the plurality of battery cells; an upper housing coupled to an upper portion of the lower housing to accommodate the plurality of battery cells; and a plurality of parallel bus bars on an upper surface of the upper housing arranged side by side to be spaced apart from each other.
5. The sub pack according to claim 1, wherein the BMS assembly includes a plurality of BMSs and a BMS frame to surround and secure the plurality of BMSs.
6. The sub pack according to claim 4, wherein the first sub module and the second sub module are connected to each other in series by an electric connection between parallel bus bars adjacent to each other.
7. The sub pack according to claim 6, wherein a parallel connection between the plurality of battery cells in a cell group, a parallel connection between the plurality of unit modules included in the first or second sub module and the serial connection between the first and second sub modules are made by wire bonding.

8. The sub pack according to claim 3, wherein each of the first BMS and the second BMS includes a plurality of sensing terminals, the sensing terminals of the first BMS being electrically connected to the parallel bus bar of the first sub module by wire bonding, the sensing terminals of the second BMS being electrically connected to the parallel bus bar of the second sub module by wire bonding.
 9. A battery pack comprising a pair of sub packs according to claim 1 coupled to each other in mirror symmetry about a central axis parallel to a longitudinal direction of the sub packs.
 10. A vehicle, comprising the battery pack according to claim 9.
-